IT

Ch9. X. The Gaussian channel, $X_i \longrightarrow \oplus \longrightarrow Y_i$ (with Z_i entering $\oplus$ from above)

$Y_i = X_i + Z_i$, Z_i is i.i.d from Gaussian distribution with variance N.

Y_i: output, X_i: input, Z_i: noise

· That is, $Y_i = X_i + Z_i$, $Z_i \sim N(0, N)$. Further assume Z_i is indep of X_i.

Here, without further condition, the capacity of this channel may be ∞.

· Thus, we assume an average power constraints:

For any codeword $(x_1, \cdots x_n)$ transmitted over the channel, we require that

$$\frac{1}{n} \cdot \sum_{i=1}^{n} x_i^2 \leq P$$

· the additive noise in the above channel may be due to a variety of cause.

· By central theorem the Gaussian is reasonable in a large number scheme.

9.1 Gaussian channel : Definitions.

Def The information capacity of the Gaussian channel with P is

$$C = \max_{f(x):\, EX^2 \leq P} I(X,Y) \quad \circledast$$

(since Z is independent of X)

then, $I(X,Y) = h(Y) - h(Y|X) = h(Y) - h(Z|X) = h(Y) - h(Z)$

Now, $h(Z) = \frac{1}{2} \cdot \log 2\pi e N$

Also, $EY^2 = E(X+Z)^2 = EX^2 + 2EX\,EZ + EZ^2 = P + N$. ($EZ = 0$, $\because EZ = 0$)

$\therefore$ Given $EY^2 = P + N$, the entropy of $Y \leq \frac{1}{2} \cdot \log 2\pi e(P+N)$

by the fact that the normal maximizes the entropy for a given variance.

$\therefore$ By using ⊛ and the above, we have

$$I(X;Y) = h(Y) - h(Z) \leq \frac{1}{2}\log 2\pi e(P+N) - \frac{1}{2}\log 2\pi e N = \frac{1}{2}\log\left(1+\frac{P}{N}\right)$$

$\Rightarrow$ $C = \max_{EX^2 \leq P} I(X;Y) = \frac{1}{2}\log\left(1+\frac{P}{N}\right)$, where the maximum is attained when $X \sim N(0, P)$.

<u>Def</u> An (M, n) code for the Gaussian channel with P consist of the following

1. An index set $\{1, \cdots M\}$
2. An ecoding fcn $x : \{1 \cdots M\} \to X^n$ yielding codewords $X^n(1), \cdots x^n(M)$ satisfying the P, that is $\forall$ codeword $\sum_{i=1}^{n} x_i^2(w) \leq nP$, $w = 1, \cdots M$.
3. A decoding fcn $g : y^n \longrightarrow \{1, \cdots M\}$.

<u>Note</u> the arithmetic average of the probability error is defied by

$$P_e^{(n)} = \frac{1}{2^{nR}} \sum \lambda_i$$

<u>Def</u> A rate R is achievable for a Gaussian channel with P if there exists a sequence $(2^{nR}, n)$ codes with codewords satisfying the power constraint s.t the maximal probability of error $\lambda^{(n)} \to 0$.

<u>9.2</u> The supremum of the achievable rates for the Gaussian chanel.

- $R \leq C$ (goal)

Pf) We must show that if $P_e^{(n)} \to 0$ for a sequence $(2^{nR}, n)$ codes for a Gaussian chanel with P $\Rightarrow$ $R \leq C = \frac{1}{2}\cdot\log\left(1+\frac{P}{N}\right)$.

Consider any $(2^{nR}, n)$ code that satisfies the P.C,

$\frac{1}{n}\sum_{i=1}^{n} x_i^2(w) \leq P$ $(1 \leq w \leq 2^{nR})$ and let W be distributed uniformly over $\{1, \cdots 2^{nR}\}$. Then,

- the uniformly distribution $W \in \{1 \cdots 2^{nR}\}$ $\Rightarrow$ a distribution on the input code words $\Rightarrow$ a distribution over the input alphabet.

$$W \to X^n(w) \longrightarrow Y^n \longrightarrow \hat{w}.$$

use Fano's inequality to get $H(w|\hat{w}) \le 1 + nR \cdot P_e^{(n)} = n\varepsilon_n$, where $\varepsilon_n \to 0$ as $P_e^{(n)} \to 0$.

$$\therefore nR = H(w) = I(w;\hat{w}) + H(w|\hat{w}) \le I(w;\hat{w}) + n\varepsilon_n = h(Y^n) - h(Y^n|X_n) + n\varepsilon^n$$

$$\le \sum_{i=1}^{n} h(Y_i) - h(Z^n) + n\varepsilon_n = \sum_{i=1}^{n} h(Y_i) - \sum_{i=1}^{n} h(Z_i) + n\varepsilon_n$$

$$(*)' \quad = \sum_{i=1}^{n} I(X_i;Y_i) + n\varepsilon_n.$$

Next, let P_i be the average power of the i-th column of the code

$\left(P_i = \frac{1}{2^{nR}} \sum_w x_i^2(w)\right)$. Note that $EY_i^2 = P_i + N$.

$\therefore$ since entropy is maximized by the normal distribution, we have that

$h(Y_i) \le \frac{1}{2} \cdot \log 2\pi e(P_i+N) \sim$ (D)

By combining (D) with $(*)'$ $\Rightarrow nR \le \sum(h(Y_i) - h(Z_i)) + n\varepsilon_n$

$$\le \sum\left(\frac{1}{2}\log(2\pi e(P_i+N)) - \frac{1}{2}\log 2\pi e N\right) + n\varepsilon_n = \sum \frac{1}{2}\log\left(1+\frac{P_i}{N}\right) + n\varepsilon_n,$$

From $\frac{1}{n}\sum_i P_i \le P$ and $f(x) = \frac{1}{2}\log(1+x)$ is concave, it follows that

$$\frac{1}{n} \cdot \sum_{i=1}^{n} \frac{1}{2}\log\left(1+\frac{P_i}{N}\right) \le \frac{1}{2} \cdot \log\left(1 + \frac{1}{n}\sum_{i=1}^{n}\frac{P_i}{N}\right) \le \frac{1}{2}\log\left(1+\frac{P}{N}\right).$$

$\therefore R \le \frac{1}{2} \cdot \log\left(1+\frac{P}{N}\right) + \varepsilon_n$. $\therefore$ we have the desired result by $\varepsilon_n \to 0$

9.3 Bandlimited channel

Def Bandlimited channel with white noise is a continuous time channel

s.t. $Y(t) = (X(t) + Z(t)) * h(t)$. → impulse response

$Y(t)$: output; $X(t)$: signal wave form; $Z(t)$ → wave form of white Gaussian channel.

Goal: How to calculate the capacity of such a channel?

To be precise, how to show that sampling a bandlimited signal at a sampling rate $\frac{1}{2W}$ is sufficient to reconstruct the signal from the samples.

thm Suppose that a fcn $f(t)$ is bandlimited to W, namely, the spectrum (Fourier transform) of the fcn is 0 for all frequencies (w) greater than W

$\Rightarrow$ The fcn is completely determined by samples of the fcn spaced $\frac{1}{2W}$ seconds apart.

pf) The proof in book is wrong.

Let $F(w)$ be the Fourier transform of $f(t)$. $\Rightarrow f(t) = \frac{1}{2\pi}\int_{-2\pi W}^{2\pi W} F\left(\frac{w}{2\pi}\right)\cdot e^{iwt}\,dw$

($\because F\left(\frac{w}{2\pi}\right)$ is 0 outsider the band $-2\pi W \leq w \leq 2\pi W$)

$\therefore f\left(\frac{n}{2W}\right) = \frac{1}{2\pi}\int_{-2\pi W}^{2\pi W} F\left(\frac{w}{2\pi}\right)\cdot e^{\frac{iwn}{2\pi}}\,dw \quad (n \in \mathbb{Z})$

The sample values $f\left(\frac{n}{2W}\right)$ determine the Fourier coefficients and by extension they determine the value of $F(w)$ in the interval $(-W, W)$.
Since a fcn is uniquely determined by its Fourier transform and since $F(w)$ is 0 outside the band W, we can determine the fcn from the samples (uniquely). Now, consider

(A) $\text{sinc}(t) = \frac{\sin(2\pi W t)}{2\pi W t}$ $\Rightarrow$ the spectrum of this fcn is constant in $(-W, W)$ and is zero outside this band.

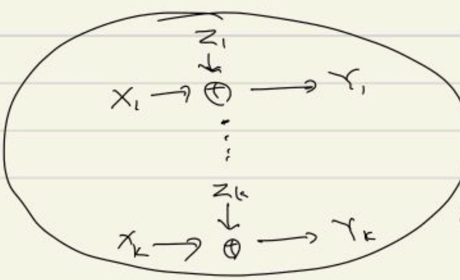
(Refer to Rudin's book. ch 10)

Now, define $g(t) = \sum_{n=-\infty}^{\infty} f\left(\frac{n}{2W}\right)\text{sinc}\left(t - \frac{n}{2W}\right)$.

$\Rightarrow$ It follows that $g(t)$ is bandlimited to W and is equal to $f\left(\frac{n}{2W}\right)$ at $t = \frac{n}{2W}$. Thus, by the (A) we get $f(t) = g(t)$.

9.4 parallel Gaussian channel

- Consider k independent Gaussian channels in parallel with a common P.
- Goal: how to distribute the total power among the channels so as to maximize the capacity with nonwhite Gaussian noise where each parallel component represent a different frequency.



First, assume that we have a set of Gaussian channels:

$Y_j = X_j + Z_j, \quad j = 1, \cdots k$ with $Z_j \sim N(0, N_j)$

and noise is independent from channel to channel.

given $E\sum_{j=1}^{k} X_j^2 \leq P$.

- the information capacity of the channel $C = \max_{\substack{f(x_1 \ldots x_k) : \\ \sum E x_i^2 \le P}} I(X_1, \ldots X_k ; Y_1 \ldots Y_k)$

- Since $Z_1 \ldots Z_k$ are independent, $I(X_1, \ldots X_k ; Y_1 \ldots Y_k)$

$= h(Y_1 \ldots Y_k) - h(Y_1 \ldots Y_k | X_1 \ldots X_k) = h(Y_1 \ldots Y_k) - h(Z_1 \ldots Z_k | X_1 \ldots X_k)$

$= h(Y_1 \ldots Y_k) - h(Z_1 \ldots Z_k) = h(Y_1 \ldots Y_k) - \sum_i h(Z_i) \le$

$\sum_i (h(Y_i) - h(Z_i)) \le \sum_i \frac{1}{2} \cdot \log(1 + \frac{P_i}{N_i})$, where $P_i = E X_i^2$

and $\sum P_i = P$. (Equality holds when $(X_1 \ldots X_k) \sim N\left(0, \begin{bmatrix} P_1 & & 0 \\ & \ddots & \\ 0 & & P_k \end{bmatrix}\right)$

$\Downarrow$ our problem is converted to

maximize $\frac{1}{2} \cdot \sum_i \log(1 + \frac{P_i}{N_i})$ given $\sum P_i = P$.

Use KKT condition to obtain $P_i = (\nu - N_i)^+$ under $\sum (\nu - N_i)^+ = P$

9.5 Channels with colored Gaussian noise.

Goal: we will consider the case when the noise is dependent. To be precise, we can consider a block n-consecutive uses of the channel as n channels in parallel with dependent noise.

- Let K_Z be the covariance matrix of the noise
  K_X be the 〃 〃 input.
- the power constraint : $\frac{1}{n} \cdot \sum E X_i^2 \le P \Leftrightarrow \frac{1}{n} \cdot tr(K_X) \le P$.
- $I(X_1 \ldots X_n ; Y_1 \ldots Y_n) = h(Y_1 \ldots Y_n) - h(Z_1 \ldots Z_n)$.

  ($\because$ determined only by their distributions, not distribution of input)

$\therefore$ We hope to maximize $h(Y_1 \ldots Y_n)$

For this, recall the entropy is maximized when Y is normal, which is achieved when the input is normal. Also, $K_Y = K_X + K_Z$ and the entropy $h(Y_1 \ldots Y_n) = \frac{1}{2} \cdot \log\left((2\pi e)^n |K_X + K_Z|\right)$, $|\cdot|$ is det.

$\Downarrow$

$\therefore$ our problem is reduced to

$\Downarrow$

choosing K_X given $\frac{1}{n} \cdot tr(K_X) \leq P$,

$\Downarrow$ $K_Z = Q \Lambda Q^T$ → diagonal matrix

since $Q \cdot Q^T = I \Rightarrow |K_X + K_Z| = |K_X + Q\Lambda Q^T| = |Q||Q^T K_X Q + \Lambda||Q^T|$

$= |Q^T K_X Q + \Lambda| = |A + \Lambda|$, where $A = Q^T K_X Q$ and then

$tr(A) = tr(Q^T K_X Q) = tr(Q^T Q K_X) = Tr(K_X)$.

$\Downarrow$ $\therefore$ The above problem is reduced to

maximizing $|A + \Lambda|$ subject to $tr(A) \leq nP$.

By Hadamard's inequality, we have that

$|A + \Lambda| \leq \prod_i K_{ii}$, with equality holds $\Leftrightarrow$ $A + \Lambda$ is diagonal).

$\therefore |A + \Lambda| \leq \prod_i (A_{ii} + \lambda_i)$, $\Lambda = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$, with equality

holds $\Leftrightarrow$ A is diagonal. Power constraint $\Leftrightarrow \frac{1}{n} \cdot \sum_i A_{ii} \leq P$, $A_{ii} \geq 0$

Use KKT condition to yield $A_{ii} = (\nu - \lambda_i)^+$, where ν is chosen so that $\sum A_{ii} = nP$.

$\therefore$ Thus, this value A maximizes the entropy of Y and hence, the mutual information //.

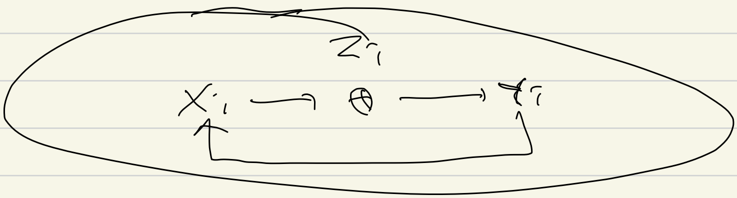
9.6 Gaussian channel with feedback (FB)

- We consider channels with memory when the noise is correlated from time instant to time instant. Feedback does increase capacity.
- We describe an expression for the capacity in terms of the covariance matrix of the noise Z.

Goal: ① To prove a converse for this expression for capacity
② simple bound on the increase in capacity due to FB.

- the Gaussian channel with FB is $Y_i = X_i + Z_i$, $Z_i \sim N(0, K_Z^{(n)})$

Note: A $(2^{nR}, n)$ for the Gaussian channel with FB consist of a sequence of mappings $X_i(W, Y^{i-1})$, where $W \in \{1, \cdots 2^{nR}\}$ is the input message and Y^{i-1} is a sequence of past values of the output.



- A power constraint: $E\left[\frac{1}{n} \sum_{i=1}^{n} X_i^2(W, Y^{i-1})\right] \le P$, $1 \le W \le 2^{nR}$
where the expectation is over all possible noise sequence

Caution: X^n and Z^n are not independent and X_i depends on the past values of Z.

- FB, without FB.

① with FB: The capacity $C_{n,FB}$ in bits per transmission of the time-varying Gaussian channel with FB is

$$C_{n,FB} = \max_{\frac{1}{n} tr(K_X^{(n)}) \le P} \frac{1}{2n} \log \frac{|K_{X+Z}^{(n)}|}{|K_Z^{(n)}|} \text{ --- } ⓐ$$

where the maximization is taken over all X^n of the form

$$X_i = \sum_{j=1}^{i-1} b_{ij} Z_j + V_i \qquad (1 \le k \le n)$$

(FB)

(innovation process) and V^n is indep of Z^n.

② with FB, The capacity C_n of the time-varying Gaussian channel with feed back is

$$C_n = \max_{\frac{1}{n}tr(K_X^{(n)}) \le P} \frac{1}{2n} \log \frac{|K_X^{(n)} + K_Z^{(n)}|}{|K_Z^{(n)}|}.$$

Thm For a Gaussian channel with FB the rate R_n for any sequence of $(2^{nR}, n)$ codes with $P_e^{(n)} \to 0$ satisfies $R_n \le C_{n,FB} + \varepsilon_n$ with $\varepsilon_n \to 0$ as $n \to \infty$, where $C_{n,FB}$ is defined as Ⓐ.

pf) Let W be uniform over 2^{nR} and use Fano's ineq to have that $H(W|\hat{W}) \le 1 + nR_n \cdot P_e^{(n)} = n\varepsilon_n$, (where $\varepsilon_n \to 0$ as $P_e^{(n)} \to 0$.

$\Rightarrow nR_n = H(W) = I(W;\hat{W}) + H(W|\hat{W}) \le I(W;\hat{W}) + n\varepsilon_n$

$\le I(W;Y^n) + n\varepsilon_n = \sum I(W;Y_i|Y^{i-1}) + n\varepsilon_n$

(by data processing ineq in ch 2)

$= \sum \left(h(Y_i|Y^{i-1}) - h(Y_i|W, Y^{i-1}, X_i, X^{i-1}, Z^{i-1}) \right) + n\varepsilon_n$

$= \sum \left(h(Y_i|Y^{i-1}) - h(Z_i|Z^{i-1}) \right) + n\varepsilon_n = h(Y^n) - h(Z^n) + n\varepsilon_n$.

Then, divide n to yield $R_n \le \frac{1}{n}\left(h(Y^n) - h(Z^n)\right) + \varepsilon_n$.

$\Rightarrow \le \frac{1}{2n} \log \frac{|K_Y^{(n)}|}{|K_Z^{(n)}|} + \varepsilon_n \le C_{n,FB} + \varepsilon_n$. ////

Q how to estimate the bdd on the capacity with FB and without FB in terms of $K_X^{(n)}, K_Z^{(n)}$??

Lemma 1 Let X and Z be n-dimensional random vectors

$\Rightarrow K_{X+Z} + K_{X-Z} = 2K_X + 2K_Z$.

Lemma 2 For $n\times n$, $A, B \succeq 0$, if $A - B \preceq 0$

$\Rightarrow |A| \leq |B|$.

Lemma 3 For $n\times n$ $X, Z \succeq 0 \Rightarrow |K_{X+Z}| \leq 2^n \cdot |K_X + K_Z|$.

Lemma 4 For $n\times n$ $A, B \succeq 0$, and $0 \leq \lambda \leq 1$

$\Rightarrow |\lambda A + (1-\lambda) B| \geq |A|^{\lambda} \cdot |B|^{1-\lambda}$.

Def X^n is causally related to Z^n if
$f(x^n, z^n) = f(z^n) \cdot \prod_{i=1}^{n} f(x_i \mid x^{i-1}, z^{i-1})$.

Lemma 5 If X^n and Z^n are causally related $\Rightarrow$
$h(X^n - Z^n) \geq h(Z^n)$.

Thm $C_{n,FB} \leq C_n + \frac{1}{2}$.

pf) Combining all the results, we have

$$C_{n,FB} = \max_{tr(K_X) \leq np} \frac{1}{2n} \log \frac{|K_Y|}{|K_Z|} \underset{\text{(Lemma 3)}}{\leq} \max_{tr(K_X) \leq np} \frac{2^n |K_X + K_Z|}{|K_Z|}$$

$$= \max_{tr(K_X) \leq np} \frac{1}{2n} \cdot \log \frac{|K_X + K_Z|}{|K_Z|} + \frac{1}{2} \leq C_n + \frac{1}{2}. \quad \boxed{Q.E.D.}$$

$\hookrightarrow$ (def of capacity with FB).

Thm $C_{n,FB} \leq 2 \cdot C_n$. $\Leftrightarrow \frac{1}{2} \cdot C_{n,FB} \leq C_n$

pf) WTS $\frac{1}{2} \frac{1}{2n} \log \frac{|K_X + K_Z|}{|K_Z|} \leq \frac{1}{2n} \log \frac{|K_X + K_Z|}{|K_Z|}$

Note that
$$\frac{1}{2n} \log \frac{|K_X + K_Z|}{|K_Z|} \underset{\text{(First Lemma)}}{=} \frac{1}{2n} \log \frac{|\frac{1}{2} K_{X+Z} + \frac{1}{2} \cdot K_{X-Z}|}{|K_Z|}$$

$$\underset{\text{(Lemma 4)}}{\geq} \frac{1}{2n} \cdot \log \frac{|K_{X+Z}|^{\frac{1}{2}} |K_{X-Z}|^{\frac{1}{2}}}{|K_Z|} \underset{\text{(Lemma 5)}}{\geq} \frac{1}{2n} \log \frac{|K_{X+Z}|^{\frac{1}{2}} \cdot |K_Z|^{\frac{1}{2}}}{|K_Z|} = \frac{1}{2} \cdot \frac{1}{2n} \cdot \log \frac{|K_{X+Z}|}{|K_Z|},$$

we complete this part.

Thank You